OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll NO. : 2520030437

Name: Bandla Vinay

SEC-8

SKILL-1:

Aim: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
admin2007@DESKTOP-MPC8TJM:/mnt/c/Users/Admin$ cd /mnt/c/V-Max_OSSP/2520030437_Skill/Skill-1
chmod +x run.sh
./run.sh
===================================================
 Compiling Skill-1 with GCC...
===================================================
Compilation successful!
Launching process_demo...
===================================================
====================================================
   OSSP Skill-1: Process Abstraction & Management
====================================================
1. Basic fork() & Parent-Child Relationship
2. exec() Family (execvp execution)
3. Process Tree Inspection
4. Run All Demonstrations
0. Exit
----------------------------------------------------

Select an option (0-4): 1

=== Fork and Parent-Child Relationship Demo ===
[PARENT INITIAL] PID: 744, PPID: 735
[PARENT PROCESS] Forked child with PID: 749. Waiting for completion...
[CHILD PROCESS] PID: 749, Parent PID (PPID): 744
[CHILD PROCESS] Child execution complete. Exiting with code 42.
[PARENT PROCESS] Child exited normally with code: 42

Select an option (0-4): 2

=== Exec Family Demonstration (execvp) ===
[PARENT] Forking child to execute 'uname -a'...
[CHILD PID: 752] Replacing image with uname using execvp...
Linux DESKTOP-MPC8TJM 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/L
inux
[PARENT] Child exec completed with status: 0
```

```
Select an option (0-4): 3

=== Process Tree Inspection (ps) ===
[PARENT] Current PID: 744
[CHILD PID: 753] Inspecting process hierarchy...
    PID    PPID COMMAND
    719     718 bash
    735     719 run.sh
    744     735 process_demo
    753     744 ps

Select an option (0-4): 4

=== Fork and Parent-Child Relationship Demo ===
[PARENT INITIAL] PID: 744, PPID: 735
[PARENT PROCESS] Forked child with PID: 754. Waiting for completion...
[CHILD PROCESS] PID: 754, Parent PID (PPID): 744
[CHILD PROCESS] Child execution complete. Exiting with code 42.
[PARENT PROCESS] Child exited normally with code: 42

=== Exec Family Demonstration (execvp) ===
[PARENT] Forking child to execute 'uname -a'...
[CHILD PID: 755] Replacing image with uname using execvp...
Linux DESKTOP-MPC8TJM 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/L
inux
[PARENT] Child exec completed with status: 0

=== Process Tree Inspection (ps) ===
[PARENT] Current PID: 744
[CHILD PID: 756] Inspecting process hierarchy...
    PID    PPID COMMAND
    719     718 bash
    735     719 run.sh
    744     735 process_demo
    756     744 ps

Select an option (0-4): 0
```

```
=== Process Tree Inspection (ps) ===
[PARENT] Current PID: 744
[CHILD PID: 756] Inspecting process hierarchy...
    PID    PPID COMMAND
    719     718 bash
    735     719 run.sh
    744     735 process_demo
    756     744 ps

Select an option (0-4): 0
Exiting Skill-1 Process Demo.
admin2007@DESKTOP-MPC8TJM:/mnt/c/V-Max_OSSP/2520030437_Skill/Skill-1$
```

Code:

```c
#include "../include/process_demo.h"
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main(int argc, char *argv[]) {
    if (argc > 1) {
        char *cmd_args[64];
        int i;
        for (i = 1; i < argc && i < 63; i++) {
            cmd_args[i - 1] = argv[i];
        }
        cmd_args[i - 1] = NULL;
        demo_custom_command(cmd_args[0], cmd_args);
        return 0;
    }

    printf("====================================================\n");
    printf("   OSSP Skill-1: Process Abstraction & Management   \n");
    printf("====================================================\n");
    printf("1. Basic fork() & Parent-Child Relationship\n");
    printf("2. exec() Family (execvp execution)\n");
    printf("3. Process Tree Inspection\n");
    printf("4. Run All Demonstrations\n");
    printf("0. Exit\n");
    printf("----------------------------------------------------\n");

    int choice = -1;
    while (choice != 0) {
```

```c
while (choice != 0) {
    printf("\nSelect an option (0-4): ");
    fflush(stdout);
    if (scanf("%d", &choice) != 1) {
        break;
    }

    switch (choice) {
        case 1:
            demo_fork_basics();
            break;
        case 2:
            demo_exec_family();
            break;
        case 3:
            demo_process_tree();
            break;
        case 4:
            demo_fork_basics();
            demo_exec_family();
            demo_process_tree();
            break;
        case 0:
            printf("Exiting Skill-1 Process Demo.\n");
            break;
        default:
            printf("Invalid option. Please enter 0-4.\n");
            break;
```